Using ecological niche modelling for population mapping and census sizes estimations of a rare plant

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Abstract

Aim) Species distribution models (SDMs) are rarely field verified at landscape scales, and it remains unclear how accurately their predictions, especially at fine spatial resolutions, reflect reality, or can guide the detection of new populations and sub-populations of rare plants. Further, their ability to extend beyond habitat classification to support estimates of population census size is unknown. We tested whether SDMs built at different resolutions, combined with distance-based sampling and density modelling, can achieve both. Location) South-central Rocky Mountains, Colorado, U.S.A. Methods) We compiled occurrence records for Eriogonum coloradense, a geographically narrow endemic, from regional herbaria and iNaturalist, and built random forest ENMs at three spatial resolutions (3, 1, and 1/3 arc-second). Before evaluating any model, we ground-truthed the raw occurrence dataset itself and conducted preliminary surveys to estimate what a dedicated survey team would document. We then tested model predictions using a spatially balanced, adaptive occupancy-based sampling design, classifying newly discovered sites as extensions of known populations, new sub-populations, or new populations based on their Euclidean and least-cost distance from existing occurrences. Plant density was modelled using both Bayesian and five machine-learning approaches (XGBoost, light gradient boosting) under spatial and non-spatial cross-validation and combined with modelled population boundaries to estimate census size. Results) Main Conclusions)

1 | Introduction

https://onlinelibrary.wiley.com/doi/10.1111/ele.12189 https://onlinelibrary.wiley.com/doi/full/10.1111/ddi.13795 # maybe https://www.cell.com/trends/ecology-evolution/fulltext/S0169-5347(18)30304-5 https://www.nature.com/articles/s41559-026-03125-y # maybe

The effects of anthropogenic stressors have led to a global biodiversity crisis, with estimates suggesting that 20-45% of plant species are facing extinction (Bachman et al. 2024; Brummitt et al. 2015; Pimm and Joppa 2015; Lughadha et al. 2020). Determining which plant species to focus our conservation efforts (e.g., active restoration, preserve creation, *ex situ* collections) on requires an array of natural history data seldom available to decision-makers (Heywood 2017). The environmental niche of a species, its occupied geographic extent, the occupancy of sites within the range, and the census size of populations are all essential attributes for evaluating a species' continued existence against anthropogenic stressors (Commission 2001; Faber-Langendoen et al. 2012; USFWS 2016). Although these attributes are relatively simple to characterise, implementation is time-consuming, requiring extensive fieldwork - hence requiring proxies for conservation assessments (Juffe-Bignoli et al. 2016; Bland et al. 2015; Pelletier et al. 2018). Environmental niche models (ENMs or Species Distribution Models, SDMs) have made enormous contributions to two of these attributes; describing the realised environmental niche of a species, and the species' geographic extent within it (Syfert et al. 2014; R. P. Anderson 2023; Kass et al. 2021). Further, this progress - paired with a growing volume of occurrence data from citizen science platforms and the digitisation of natural history museum records - offers promise for modelling the remaining two attributes, occupancy and census size (Markham et al. 2023; Feldman et al. 2021).

However, despite this progress, the utility of models is seldom field verified, especially at landscape scales (Chiffard et al. 2020; A. Lee-Yaw et al. 2022). Even despite an awareness that ENMs generally suffer from having few, often spatially biased, occurrence records as dependent variables, which often fail to characterise the ecological breadth of the species niche (Stolar and Nielsen 2015; Feeley and Silman 2011; Pili et al. 2025). Accordingly, while many ENMs achieve high performance on hold-out test subsets, biases in the training data may be propagated into predictions (A. Lee-Yaw et al. 2022). Increasingly adaptive-niche-based sampling (ANBS), which generates larger data sets of both presence and absence records for training – leading to more robust models, while possibly obtaining more data from sparsely sampled niche space. However, most ANBS sampling has preferentially visited cells with high predicted occurrence probability (Guisan et al. 2006); because such cells are often close to already-known occurrences, this tests a model's easiest predictions rather than expanding its training data to better reflect the species niche. Refitting a

model with this newly acquired data not only expands the presence/absence pool ((ALLOWING BETTER PARTINIONING OF CONTRIBUTIONS OF INDEPENDENT VARIABLES)), but also verifies coordinate placement, confirms that historic points are still extant ((BOTH DECREASING RECORDS OUTSIDE THE NICHE)), reduces model uncertainty, and may yield additional data such as census estimates and life stage demographics (Stockwell and Peterson 2002; Wisz et al. 2008; Jansen et al. 2022; Mondain-Monval et al. 2024). Ground-truthing results are, in any case, seldom shared or reported for plant species, and even less effort has apparently been made to first verify the baseline reliability of existing records for rare species specifically (Johnson et al. 2023; Borokini et al. 2023; Zimmer et al. 2023).

The spatial grain of environmental variables has often been too coarse to model the micro-environments that govern many rare plants' distributions, especially in heterogeneous environments (Guisan et al. 2013). Accordingly, most early ENM work targeted resolutions useful for conservation policy decisions, while resolutions fine enough for natural resource professionals' individual management decisions remain nascent. Recent advances in remote sensing, statistics, and software have narrowed this resolution mismatch (Zellweger et al. 2019). Recent papers have had mixed results regarding the effects of imprecisely mapped occurrence data on ENM predictions, with indications that models generated in more heterogeneous environments, and at finer resolutions (e.g. ca. 3 arc-second to 10 arc-minutes (~14.5km at 38°)) suffer minor decreases in model performance (Graham et al. 2008; Smith et al. 2023) with real species, while increasing error in mapping has drastic effects on model predictions with virtually simulated species (Gábor et al. 2022).

An ENM predicts a single variable: the probability of suitable habitat for the taxon (taddeo2026diversitydistributions). However, in most conservation applications, the insight desired from an ENM is actually the species' realised distribution (cite). Significant progress has been made integrating ENM predictions with models of dispersal, recognising that the bridge between an ENM and a plant population's presence is propagule dispersal and establishment, not the distribution of the fundamental niche alone (Guisan and Thuiller 2005; Franklin 2010; Engler and Guisan 2009). Previous innovations have largely focused on connecting currently suitable habitat to habitat predicted to be suitable under climate change (Engler and Guisan 2009; Bocedi et al. 2014), rather than on detecting populations that already exist but remain undiscovered. To assist practitioners in detecting new populations or extending the range of known populations, we propose modelling plant occupancy as a random variable that depends on distance from known occupied sites. 'Distance' may be defined as Euclidean (or Haversine), or as a least-cost distance reflecting a generalised surface which conveys the difficulty for propagules (e.g. seeds) to travel between the nearest occupied sites and the site of interest (Etherington 2016). By combining habitat suitability with both Euclidean and least-cost distance, we distinguish candidate sites as extensions of known populations, new sub-populations,

or new populations, rather than relying on suitability alone to prioritise field verification.

Obtaining reliable estimates of plant population census sizes requires two pieces of information: 1) measurements of plant density and 2) population extent. Observers often estimate population sizes rather than directly measuring boundaries or density, which can be unreliable and does not carry estimates of uncertainty (Reisch et al. 2018). Several field methods for acquiring density measurements have been developed (Rominger and Meyer 2019; Krening et al. 2021; Ermakova et al. 2021; Alfaro-Saiz et al. 2019; Schorr 2013) to estimate population census size (N), as well as genomic methods to estimate effective population sizes (N_e) (e.g. linkage disequilibrium) (Waples 2024). However, these methods are focused on descriptive rather than predictive processes, estimating uncertainty in measurements *within* a population rather than across the species' range without the use of covariates. We propose generating density estimates with spatial statistical models that predict both estimates and uncertainty across gridded surfaces (Oliver et al. 2012; Doser et al. 2024; A. Lee-Yaw et al. 2022). The second requirement, mapping a population's boundaries, is similarly time-consuming but essential for an accurate census size estimate. Currently population boundaries are delineated implicitly: practitioners walk outward from the population until gene flow between individuals should be minimal (e.g., 1km), searching for additional individuals, or rely on aerial imagery (Elzinga et al. 1998). Many rare plant species are small and inconspicuous, and many taxa grow in small, clustered groups in specific habitats, making detection via either method difficult (G. Chen et al. 2013; Condit et al. 2000).

[Figure 1 about here.]

1) An ENM models the continuous probability of suitable habitat for a taxons realized niche, that is the habitat it is expected to grow to maturity in, based on known occurrences.
2) The ENMs continuous probability function is converted into a presence / absence flag using thresholds along the gradient. The threshold may try to reduce false positives, if a conservative range estimate is required, reduce false negatives for a more inclusive result, or balance these characteristics.
3) Suitable habitat in closer proximity to known occurrences is more likely to be populated, and indeed the lack of notation may simply be due to the time required to map presences, a form of the beach problem.
4) Plant density is modelled using a similar suite of covariates as an ENM, with the option to include the ENM as the variate.
5) Census size estimates are a combination of plant density throughout a population, and a populations border.

Here we: 1) evaluate the results of the best-performing model with ground truthing (within 270m of existing

occurrences). 2) Compare the results of models from different resolutions to the best model, as based on hold-out test data. 3) Determine whether model predictions are useful to find new sub-populations (270m - 450m from existing occurrences), or populations (> 450m from existing occurrences). 4) Estimate population census size using 4a) estimates of plant density, and 4b) modelling of population boundaries.

2 | Methods

2.1 | Study Species & System

Eriogonum coloradense Small (Polygonaceae) is a synoecious, mat-forming perennial herb endemic to the south-central Rocky Mountains of Colorado, U.S.A. Its known range spans five counties and encompasses 22 documented populations, occurring across a broad range of elevation (2700-3900m asl), slope, aspects, soil types, and habitats - from high-elevation sagebrush steppe, through subalpine grasslands, to alpine slopes. It is ranked S3/G3 by NatureServe and listed as a Tier 2 species in Colorado Parks and Wildlife's State Wildlife Action Plan [D. G. Anderson (2004); NATURESERVE].

This taxon is largely pollinated by solitary bees, flies and butterflies, which are known to generally forage under 1km(Gathmann and Tscharntke 2002; Zurbuchen et al. 2010). Further given the topographic roughness of this area, we tenatively define populations as being isolated by <500 meters, translated to 470m to match the 90m resolution grain.

2.2 | Data Acquisition

Occurrence records were downloaded from iNaturalist (80 records) and the Consortium of Southern Rockies herbaria (131 records) (iNaturalist 2024; *Southern Rocky Mountain Herbaria Portal*, n.d.). These data were manually reviewed, and 16 herbarium records were removed due to low geolocation quality or a georeferenced locality inconsistent with the specimen label which we could not accurately identify; leading to 107 presence records.

Digital elevation models (DEMs) at 3 arc (TYPE) and 1 arc (TYPE) resolution, and digital elevation products (DEPs) at 1/3 arc resolution, were acquired from the United States Geological Survey and clipped to the analytical domain, a rectangle buffered 16 km (10 mi) beyond any known population. These elevation products were used to derive most geomorphology datasets with whiteboxTools (Wu and Brown 2022), plus heat load and diurnal anisotropic heat indices calculated with spatialEco (Evans and Murphy 2025).

Vegetation cover data were generated by combining raster layers into three continuous cover classes: Forested, Shrub, and Herbaceous (Tuanmu and Jetz 2014).

ClimateNA was used to downscale GridMet (Cite) data to 3 arc-second resolution, which was then bilinearly interpolated to generate products at the 1 and 1/3 arc resolutions (Wang et al. 2016). Gray-Level Co-occurrence Matrices (GLCM) were produced with the glcm R package from 2023 NAIP aerial imagery (CITE), bilinearly resampled to the original resolution, using default settings with a window size of 5 in both directions (Zvoleff 2020).

Collectively these resulted in 50 independent variables for ENM modelling.

2.3 | Ground Verification (H1)

As our taxon has not had recent targeted survey work carried out for a it, a first round of sampling was used to detect occurrences which would be found by a field botanist without the need of statistical modelling approaches and thereby inflate the results of our study. The first round of ground sampling was carried out from June to September 2024. All pre-existing occurrence points were considered candidates for revisits, and 25 trails leading to them were marked as sample areas. Each trail was manually mapped and buffered by 45m in each direction; 551 random points were drawn, thinned to distances $>$ 90m, leaving 309 random plots for assessment. 13 trails were visited, allowing for the assessment of 171 random points, of which 20 had new occurrence records. When conducting fieldwork presences of *E. coloradense* were opportunistically noted to better describe the spread of the population, quadrats were quasi-randomly placed - accounting for local abundance - ca. 30-50m from the previous plot until traversing out of the population (n = 246). Subjectively placed absences were also collected in areas that seemed favourable or were in proximity to *E. coloradense* individuals; this occurred in the field (n = 129). Absences in areas known to be outside the range of the populations were also added later via aerial imagery (n = 66). At each occurrence site, individuals were counted by three maturity stages (mature, juvenile, and seedling) in 3x3m quadrats.

2.4 | Comparison of Different Spatial Resolutions (H2)

Environmental niche models were generated at three spatial resolutions: 3 arc-second (~72m), 1 arc-second (~24m), and 1/3 arc-second (~8m). The modelling details followed the same process across spatial resolutions. Spatial data processing was largely performed in R (R Core Team 2026), using sf (E. Pebesma 2018) and terra (Hijmans et al. 2026), with modelling implemented with the tidymodels framework (Kuhn and Wickham 2020).

Records were thinned to the distance of a hypotenuse of a cell to avoid pseudo-replicates using spThin (Aiello-Lammens et al. 2015). 1.1 times as many background absence records as presence records were generated using the environmental distance from the sdm package (Naimi and Araujo 2016). These records were manually reviewed, and six records in areas with possible presences were removed. After this, the records were randomly sampled to reduce the dataset to XX records. After the first modelling iteration, all additional presences and absences were thinned using spThin similarly and combined with the original absence records. 'Presences' with greater coordinate uncertainty than the modelling resolution were removed.

Prior to fitting, uninformative predictor variables were removed for each model using Boruta feature selection, retaining both confirmed and tentative important variables (Kursa and Rudnicki 2010). The mTry and tree-depth hyperparameter were tuned using caret (Kuhn and Max 2008). All random forest models were fit using the ranger package; 95% confidence intervals were generated using the infinitesimal jackknife (Wager et al. 2014; Wright and Ziegler 2017).

2.5 | Occupancy Based Field Sampling (H3)

360 candidate ground-truthing sites and 180 oversamples were selected using a spatially balanced Generalized Random Tessellation Stratified (GRTS) design (Dumelle et al. 2023), stratified by named site regions and allocated proportional to each region's candidate-pixel count (DEFINED AS?), which was restricted to areas which a human could safely sample. Within each region, pixels were prioritised using a weighted combination of 1) predicted suitability, favoring the ~50% probability band; 2) Euclidean distance from known occurrences, favouring the underrepresented far tail; and 3) a cost-distance anomaly. The cost-distance anomaly utilises within-region residuals of least-cost distance (Etten 2017) regressed on Euclidean distance, which flags cells where terrain-based travel difficulty diverges most from straight-line expectations.

2.6 | Plant Density (H4a)

Plant density (plants per 3×3 m quadrat) was modelled with a ladder of approaches of increasing complexity, each evaluated under spatial and non-spatial cross-validation (Milà et al. 2022) and against the held-out 2026 field plots. Because low-elevation plots hold roughly sixfold more plants than high-elevation ones, models were fit and scored separately within each elevation phase. Within each phase, train and test sets were partitioned by data twinning on rescaled coordinates and counts (Vakayil and Joseph 2026).

We distinguish here three scenarios for turning the density surface into a census estimate, relying on the area of applicability (AOA) (discussed in Section 2.8), and whether census plots exist per each population. The

AOA includes areas within the environmental niche of the training data, while areas outside are considered extrapolation. Populations with plot data, where we seek to estimate the populations census size, are to be described; while populations within the AOA without census data are to be predicted. Under interpolation scenarios Bayesian hierarchical models are preferred as their posterior credible intervals effectively estimate uncertainty. However, spatial basis terms (used in hierarchical models) can have difficulty describing unsampled areas, and for prediction alternative models such as machine learning models may be required; although calculating estimates of uncertainty for these is time consuming. Outside the AOA (extrapolation), fitted models cannot be reliably applied, and estimates from simpler interpolators may remain valid for coarse, directional decisions.

The ladder ran from 1) null baselines, 2) interpolations, 3) fitted bayesian hierarchical models and 4) gradient boosted tree models. The nulls models were per-phase mean and median. Interpolated models were ordinary kriging (E. J. Pebesma 2004), inverse-square inverse-distance weighting, and k-nearest-neighbour averaging, and a single-covariate Poisson GLM on habitat suitability. The Bayesian hierarchical models (brms; Poisson, negative-binomial, and hurdle negative-binomial, each with population as a fixed effect and a Gaussian-process spatial term) (Bürkner 2017). The gradient-boosted trees were XGBoost with Poisson and Tweedie likelihoods, and LightGBM with Poisson likelihoods (T. Chen and Guestrin 2016; Ke et al. 2017).

2.7 | Population Boundaries (H4b)

Population boundaries were delineated by thresholding the continuous habitat suitability surface from the best-performing model using two different threshold-selection rules 1) sensitivity, and 2) equal sensitivity and specificity (Hijmans et al. 2024). Sensitivity is the ability of a model to accurately predict occurrences at the expense of false positives, while equalizing it with specificity balances accuracy of both classes (Hijmans et al. 2024). Because the sensitivity-rule polygons were nested entirely within the spec_sens-rule polygons rather than crossing them, we used the smaller (sensitivity-rule) polygon's outer edge as the baseline for transect placement, treating the band between the two polygons as the zone where the two threshold rules disagree - and the area to test which rule better represents the population's actual extent. For each patch, we identified the smaller polygon's outer edge using native raster boundary detection (4-connected neighbours), generating candidate transects along the four cardinal directions from each edge cell. Transects began 50m inside this edge, extended outward across the disagreement band until clearing the larger (spec_sens-rule) polygon's boundary by a further 15m, and were capped at 150m total length, so that each transect crosses both threshold boundaries and directly tests which rule better matches the field-observed edge of the population. Patches containing a known occurrence within 100m received a size-proportional transect budget (area^0.5,

floor of 3, ceiling of 12 per patch); patches without a nearby known occurrence were flagged as exploratory and given only a fixed floor of transects, kept as a separate sampling stage. A per-site cap of 15 transects (trimming exploratory transects first) and a no-crossing constraint were applied before finalising the field sheet, which records each transect's start coordinates, true and magnetic compass bearing, and a series of stations at 10m intervals along its length.

2.8 | Model Refinement

Additional species distribution models were fit using the iteration 0 and iteration 1 data, and indepedently tested using the second ground truth round of data. They represent refinements of modelling approaches not explored during the original iteration 1 phase, but candidates for the final modelling process. These refinements resulted from a realisation during ground truthing that two separate processes are desired of SDMs; the first to map known occurrences, the second to map suitable habitat (Pili et al. 2025). They further aimed to address 1) biases in the presence data, 2) observed variance in the effects of test/trains splits on model holdout evaluation, and 3) better understanding model extrapolation.

The accuracy of models fit in iteration 1 using the random CV strategy, at a fixed presence:absence ratio, varied widely in precision, with holdout scores varying by 10% between replicants. This variance was attributed to three possible sources: (1) the identity of the pseudo-absence draw, (2) the train–test partition (which presences fall in the test versus the training fold), and (3) genuine differences in model performance. To separate the first two sources, we introduced two spatial cross-validation strategies. Leave-multiple-populations-out (LMPO) targets prediction of habitat for population discovery, and a spatial subsetting approach (spatial mapping) targets prediction of known occupied habitat [DESCRIBED IN APPENDIX XX]. These CV strategies allow using both the same single train-test split, and the same presences in each CV fold per replicate, varying only the pseudo-absence draw allowing us to address 1. To address (2), the outer train-test split and the composition of each CV fold were allowed to vary across each PA ratio at five replicates for each spatial CV strategy. These strategies have a degree of spatial constraint in their splits, which we hypothesised w-0.119 [-0.242, -0.056] ould reduce variation in holdout variance. The third source, whether variation in accuracy was due to actual model performance was tested using fixed effects modelling of the iteration 1 classic CV strategy and results from (2), using nAP on ground truthed data as a function of heldout nAP, with effects for modelling strategies.

For overall model classification we report three complementary discrimination metrics: the Boyce index, ROC-AUC, and normalized average precision (nAP) which corrects PR-AUC across different presence to absence ratios (Flach and Kull 2015). The Boyce index measures presence-background rank-calibration, but

is insensitive to monotone rescaling and understates prediction magnitude (Hirzel et al. 2006). ROC-AUC measures discrimination, but is prevalence-invariant and so inflated by background placement rather than count (Fawcett 2006; Lobo et al. 2008; Jiménez-Valverde 2012). nAP $(\mathrm{AP} - \pi)/(1 - \pi)$ allows comparison of PR-AUC, which conveys the model's ability to discriminate the positive class, across difference prevalences (Davis and Goadrich 2006; Saito and Rehmsmeier 2015; Sofaer et al. 2019; Allouche et al. 2006).

To better characterise biases in the models we report the Cox calibration intercept and slope (Cox 1958; Steyerberg and Vergouwe 2014). The Cox intercept measures overall bias; unbiased models have values near zero, while models which predict too highly have negative values, and models with predictions too low have positive values. The Cox slope measures confidence; a correctly scaled model has values near one, overconfident models (predictions too extreme) have values below one, and and underconfident models have values above one. To reduce bias in prediction confidence, model predictions within the AOA are recalibrated using Platt scaling [*CITE*]. Platt scaling is only used within the AOA because the calibrator is fit on in-sample data, and is sensitive to extrapolation. Cox slopes from model holdouts are reported raw, while areas within the AOA are reported after calibration.

We report performance across three prediction domains defined by the area of applicability (AOA) and distance to known populations: (1) inside the AOA and within 270m of a known population, (2) inside the AOA but $>$ 270m, and (3) outside the AOA (extrapolated niche). Domain (1) is both within the models AOA and presumed reachable via dispersal, errors here are interpretable as inaccuracies of the model. Domain (2) while within the models AOA cannot be presumed reachable via dispersal, and errors are more difficult to trace directly to either model inaccuracy or dispersal liminations. Domain (3) is environmental space unsampled by the biased training data (e.g. the Antero region), where failure reflects niche truncation in the data rather than a deficiency of the model *per se*.

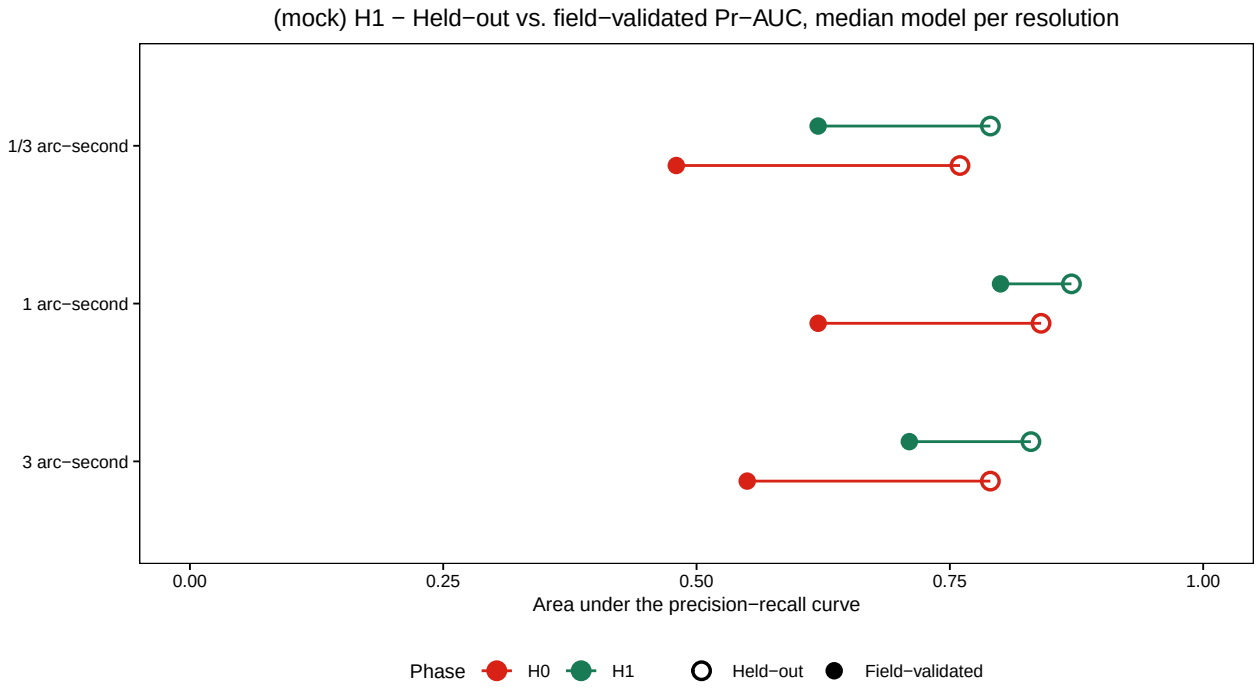
2.9 | Final Models

Using occurrence, density, and boundary-transect data from both rounds of ground-truthing combined with the original records, we refit final models of habitat suitability, occupancy, and census size, evaluating each solely against a held-out subset of the ground-truthed data not used in any prior training or tuning step.

3 | Results

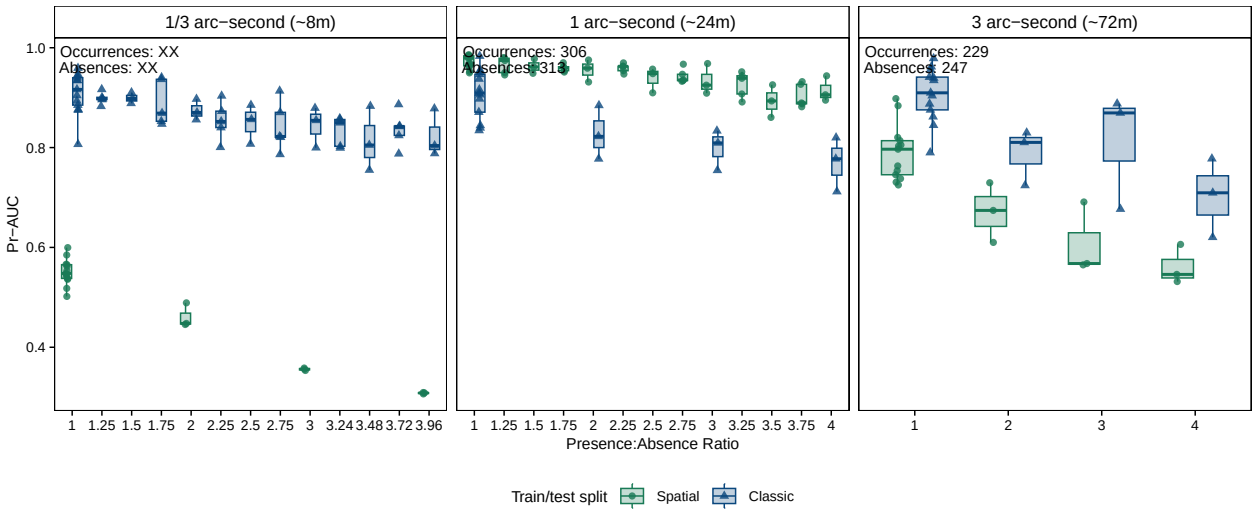
3.1 | Ground Verification (H1) // USES ONLY ITER0 and ITER1

Within 270m of the known occurrences, the 3 arc-second model (n = XX) had a PR-AUC of XX, indicating they can model the species fundamental niche. The differences between resolutions varies by XX-XX, with 3-arc second models' performance likely inflated due to a trump ruling where absences near presences are removed for evaluation.



3.2 | Comparison of Different Spatial Resolutions (H2) // USES ONLY ITER0 and ITER1

The 1/3 arc-second model (n = obs plot, n = obs opportunistic) has a PR-AUC of, 1 arc … , 3 arc. If 'removing' the points not compared in the 3arc model the 1 arc PR-AUC becomes, and removing the points not compared in the 1 and 3 arc second models the 1-3 arc model has THIS pr-auc. Across both modelling runs, the (non-)spatial CV model diagnostics were more like the ground truthed data.



3.3 | Occurrence Dectection and Occupancy Based Field Sampling (H3) // USES ONLY ITER1

Two new populations (> 450m from a known population) were detected: Cochetopa Dome - Highway 114 (based on field crews intuition) with mapped quadrats, and Antero - Spring Creek (based on iNaturalist) mapped quadrats. There were large areas of occurrences, 2 subpopulations (270-450m from a known population), and 2 major extensions (100-270m from a known population), attributable to the intepretation of the models by field workers. The two new subpoulations, a large one on Hasley Basin - Ridge South mapped quadrats, and a small one on Fridgid Ridge mapped quadrat. The two extensions occurred on both Hasley Basin - South mapped quadrats, and Crystal Peak mapped quadrat.

Across all new occurrence points detected during the 2026 modelling, less those in the Antero region, a logistic regression model treating occupancy as a dependent variable on euclidean and cost distance to known occurrence was fit. The model of occupancy on Euclidean distance and the cost-distance anomaly finds that neither predictor is significant alone (distance: $p = 0.15$; resistance: $p = 0.083$), but each becomes significant once the other is included (distance given resistance: $p = 2.7 \times 10^{-6}$; resistance given distance: $p = 1.6 \times 10^{-6}$), indicating the two distance measures are complementary rather than redundant predictors of occupancy.

3.4 | Plant Density (H4a) // USES ONLY ITER1

[Table 1 about here.]

[Table 2 about here.]

Our results show that census estimation methods are roughly comparable across interpolation scenarios. However, Bayesian and machine learning approaches allow for explicit modelling of non-zero count cells, making them an additional utility in total population sizes. Bayesian models performed well in descriptive application within the aoa ((stats)). Machine learning models, particulary XX, performed ((stats)) for predictive tasks within the AOA. Finally, extrapolation estimates may best be accomplished by using descriptive statistics outside the AOA.

3.5 | Population Boundaries (H4b)

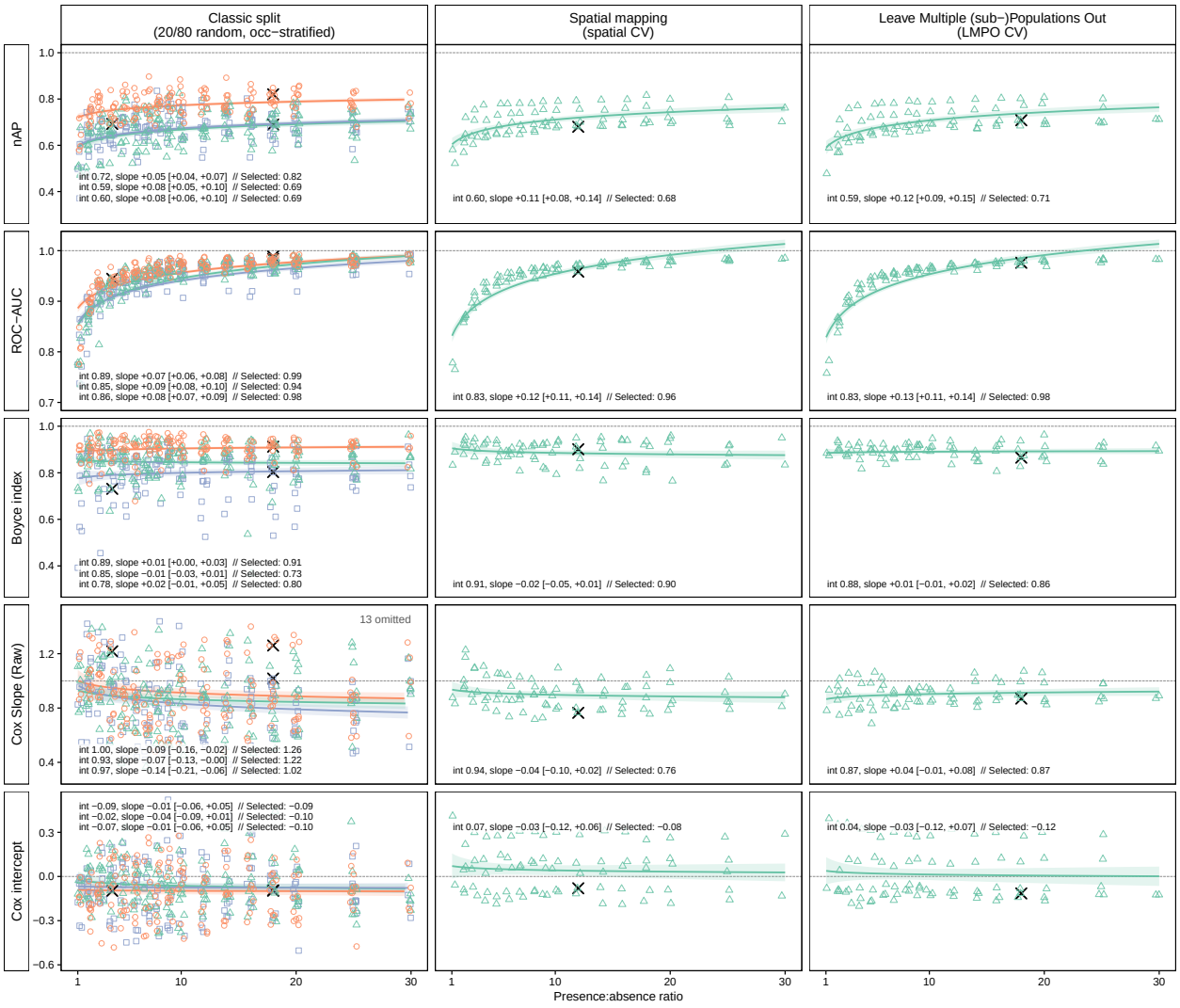
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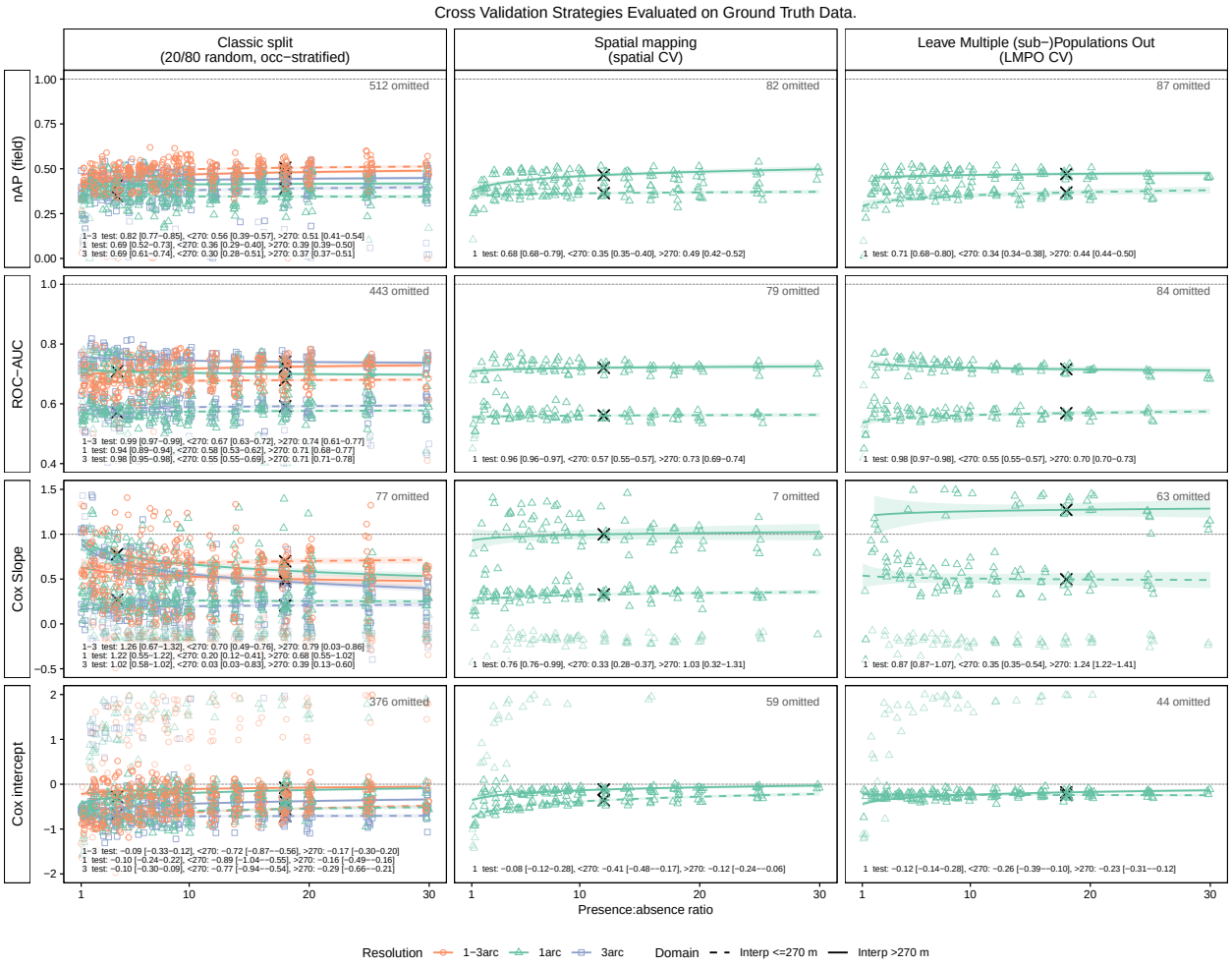
3.6 | Ground Truth Based Model Refinement

Rough idea: - determined in GT cycle that pr-AUC alone does not lead to best "on the ground model" - also pr-auc maximized at low PA ratios; while user inspection found higher PA ratios more likely (from map results) - use nAP to decrease the penalty of record prevalence on results. - added two forms to assess model bias and confidence. - maintained roc-auc to work on easy discrimination of non-suitable habitat

- Also found indications that test/train splits greatly changed model TEST accuracy results
  - suspected this not related to ACTUAL performance.
  - isolated into pseudo-absence effects via seeds for LMPO and spatial mapping
  - versus pure presence loading variation.

Cross Validation Strategies Evaluated on 'held out' Test Data.





These CV strategies allow using both the same single train-test split, and the same presences in each CV fold per replicate, varying only the pseudo-absence draw allowing us to address 1. To address (2), the outer train-test split and the composition of each CV fold were allowed to vary across each PA ratio at five replicates for each spatial CV strategy.

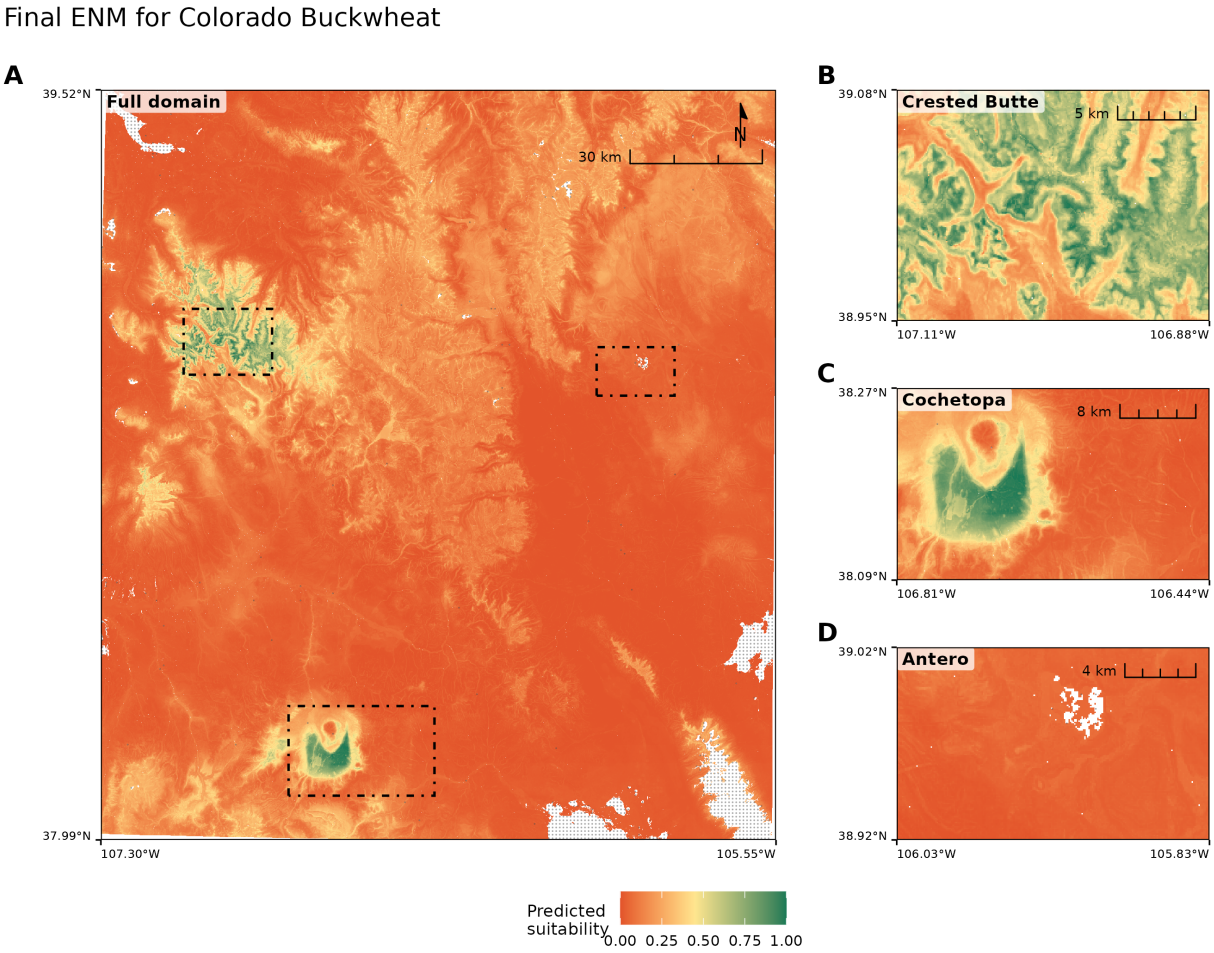
3.6.1 The effect of psuedo-absences on model evaluation Holding the train–test split and the CV-fold presences fixed and varying only the pseudo-absence draw — three draws per configuration, across both the LMPO and spatial-mapping strategies — produced negligible variation in model diagnostics across resolutions and presence:absence ratios (1:1–1:30): median within-cell standard deviation of 0.005 in nAP, 0.002 in ROC-AUC, and 0.030 in Cox slope. This insensitivity concerns the *identity* of the pseudo-absences at a fixed count; their *number* (the presence:absence ratio) remained a large source of variation (Section 3.6, above), and it pertains to a landscape background pool drawn predominantly far ($ $30 km median) from presences.

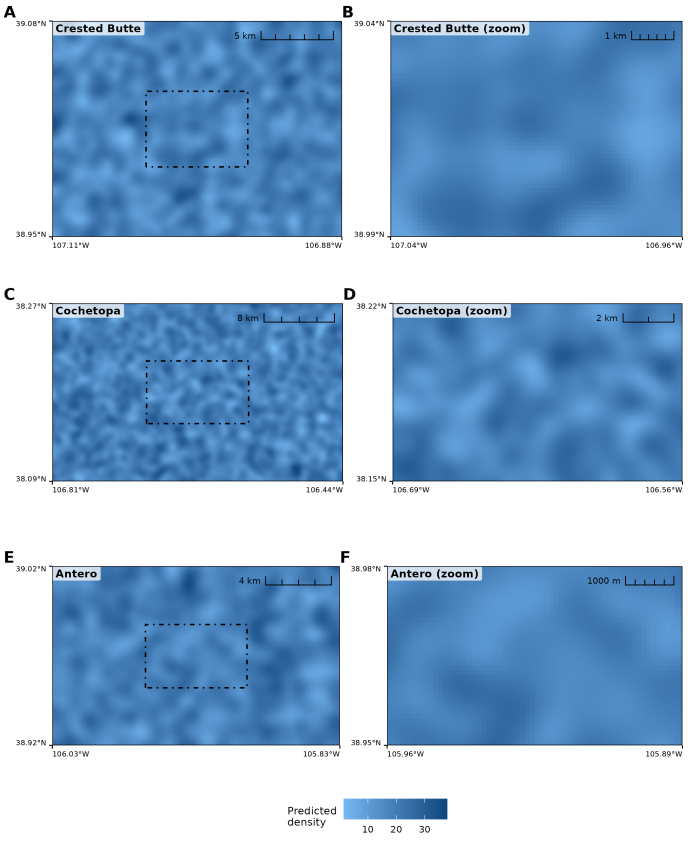
3.6.2 Whether perfomance metrics of hold-outs transferred to on the ground accuracy

Using a linear fixed-effects model (n = 406, R^2 = 0.43, adjusted R^2 = 0.42), we found strong evidence (partial R^2 = 0.06, p = 3.9×10^{-7}) of a positive, yet weak, relationship between the nAP of held-out test data and nAP of the independent ground-truth data (slope = 0.24, 95% CI 0.15–0.33). This model accounted for spatial resolution (partial R^2 = 0.13, p = 1.0×10^{-12}), prediction domain (partial R^2 = 0.15, p = 3.2×10^{-15}), and cross-validation strategy (partial R^2 = 0.03, p = 0.01). The strength of this relationship did not differ among cross-validation strategies (interaction p = 0.37).

3.6.3 The effect of different presences in train and test splits on model evaluation

3.7 | Final Results





4 | Discussion

Be sure to mention –

1) the iNaturalist records found outside the niche of the species.
    a) Relate bias of RF to this.
2) Field area was selected to best represent amount of field sampling time available to crews, which often must travel considerable distances day of sampling to reach populations.
    a) However, area is one of the best characterised alpine ecosystems globally due to its proximity to the Rocky Mountain Biological Laboratory.
3) While very high-resolution spatial data are becoming increasingly available, they may not be required for assessment of many species' niches, and existing standard of YADA YADA for naturserve is still valuable.
4) Every approach generates estimates of statistical uncertainty, albeit some (infinitesimal jackknife) are highly time intensive.

Methods to prevent the extrapolation of SDMs beyond there training data has been a major concern [thuiller2004effects]. Here, by utilising areas of applicability, based on the presence data only, we show that AOA is only a fraction of an already small geographic domain. Further, in the presence of strongly geographically biased data, it is able to highlight portions of the realized species distribution entirely. Hence these masks should always be created to refine the understanding of where a model may make inference too on the landscape.

As expected the relationship between evaluation metrics calculated on held out test data and independent performance are weak, indicating considerable variation in model performance is due to variation in test/train split partitions rather than actual model performance.

This separates the two tasks SDMs are routinely asked to perform but which biased data force into a trade-off: mapping the known, occupied niche versus characterising the fundamental niche (Thuiller et al. 2004; Jiménez-Valverde 2012; Lee-Yaw et al. 2022). The usual evidence for this trade-off is simulated (virtual species); here the 2026 field data let us observe it directly – field-verified presences in predictor space the model scored unsuitable – making the distinction between a biased model and an inadequate one empirically, rather than theoretically, visible.

5 | Conclusions

ENMs have proven themselves an essential tool for conservation planning. Recent advances in statistical and remote sensing have extended this utility to land management decisions. Here we showed that ENMs can also be elaborated upon to model other attributes essential for conservation planning and land management decisions such as census size estimates. We provide a framework for occupancy modelling and further exemplify the utility of spatial cross-validation in predictive modelling. Our work re-iterates the value of citizen science, exemplified here by the discovery of populations by three individuals on iNaturalist outside of the predictions of our biased models – and which the lead author would certainly not have visited without their observations. Finally, we hope that others build upon our data sets using an ANBS approach with this taxon, as our 'final' results are simply another hypothesis yet to be tested.

6 | Author Contributions

R.C.B.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing. S.T. and J.B.F.: Writing. All authors read, revised and approved the final manuscript.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All analysis code, and occurrence data are available on GitHub at [REPO URL]. While the individual fitted models, 3arc resolution predictors, and raster surface predictions, are archived on Zenodo [DOI].

Appendix

1) More informative predictors, or large sample sizes? How does accuracy change in light of occurrence number
2) High leverage observations influencing uncertainty in ENM predictions.
3)

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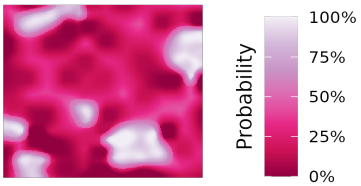
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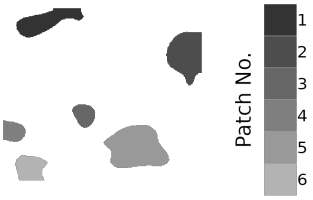
Relationships between facets of ENMs

Suitable Habitat



The probability of suitable habitat indicates areas which range from favorable (100%) to inhospitable (0%) for the species, based on the training data. Different techniques are used to interpret this probability, and to apply thresholding to it... This probability can then be used as a covariate in further modelling (e.g. density), or derived for other applications such as generating patch shapes and sizes, noting occupancy, and the distance between occupied and unoccupied patches.

Suitable Patches



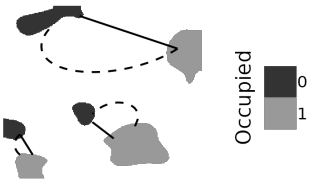
Species are limited to areas of habitat which are large enough to support positive population growth, or which serve as sinks of propagules. Given the density of populations, and size of invididual plants, not all patches have enough area to support an effective population. Further, species vary in their ability to grow on peripheral or marginal habitat around optimal habitat, decreasing the true patch size. Finally smaller patches create smaller "targets" for propagules to disperse to, decrease the probability of plant establishment.

Patch Occupancy



Training data explicitly denote the occupation status of certain areas of suitable habitat. Using these data we can mark patches of suitable habitat which contain known individuals or populations. These occupied patches can be combined with measures of distance to set up an adaptive sampling schema to find new populations.

Distances



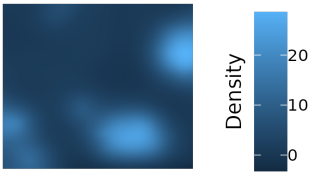
Patches are separated by both geographic (measured as euclidean for short distances or Haversine to accomodate the curvature of the earth) and environmental distances. These distances decrease the likelihood of propagules dispersing between patches. Environmental distances can be estimated by first creating generalized least-cost surfaces and then using various cost algorithms (e.g. electrical conductivity) to link patches known to be occupied with unvisited patches of suitable habitat.

Boundaries



Different thresholds of habitat probability can be used with field verification to most accurately determine where populations end. These boundaries can then be used to refine patch shapes, the distances between them, and to estimate census sizes of the populations in a final modelling process.

Plant Density



The number (abundance) of plants per specific area can be modelled and projected onto gridded surfaces. This modelling process can involve using the ENM probability surface as a covariate and clipping estimates to predefined boundaries. Summing values from all raster cells across cells can produce an estimate of the number of plants per (sub-)population.

Figure 1: figgy stardust

List of Tables

1 Field-validated mean absolute error (MAE, plants per 3×3 m quadrat) of count models against the 2026 ground-truth plots. Models are scored within each elevation phase and split by prediction regime: *interpolation* (plots inside the sampled range) versus *extrapolation* (the Antero site, 86 km beyond it; low elevation only). *n* columns score all plots (zeros included), *occ.* columns occupied plots only; '—' marks the gp()-based Bayesian models, whose spatial term extrapolates unreliably beyond the sampled range, so out-of-domain errors are not reported. 29

2 Observed plant counts per 3×3 m quadrat on occupied plots (count > 0) in each area, giving the scale against which the mean absolute errors in the preceding table should be read. 30

Table 1: Field-validated mean absolute error (MAE, plants per 3×3 m quadrat) of count models against the 2026 ground-truth plots. Models are scored within each elevation phase and split by prediction regime: *interpolation* (plots inside the sampled range) versus *extrapolation* (the Antero site, 86 km beyond it; low elevation only). *n* columns score all plots (zeros included), *occ.* columns occupied plots only; '—' marks the gp()-based Bayesian models, whose spatial term extrapolates unreliably beyond the sampled range, so out-of-domain errors are not reported.

	Interpolation				Extrapolation	
	High elevation		Low elevation		Low elevation	
Model	n = 111	occ. n = 60	n = 33	occ. n = 20	n = 33	occ. n = 16
Null models						
Arithmetic mean	4.26	3.83	37.66	41.64	40.84	50.73
Median	3.44	6.37	38.05	53.35	37.47	61.88
Interpolation & GLM baseline						
Ordinary Kriging	4.49	3.71	37.35	42.02	37.80	60.25
IDW (inverse-square)	4.98	4.03	37.69	40.91	36.90	66.99
kNN (within pop)	6.51	4.95	38.18	44.38	36.90	67.01
kNN (across pop)	4.53	3.55	38.28	55.43	36.90	67.01
GLM Poisson ~ suitability	3.83	5.06	37.84	41.74	40.55	51.40
Gradient-boosted trees						
XGB Tweedie	3.58	5.30	32.49	41.02	37.12	64.78
XGB Tweedie (Spatial CV)	3.53	5.29	31.57	48.40	36.89	73.09
XGB Poisson	3.69	5.96	34.82	54.83	36.30	74.87
XGB Poisson (Spatial CV)	3.66	5.80	35.10	54.66	36.30	74.69
LGBM Poisson	4.21	5.36	35.73	58.19	36.30	74.87
LGBM Poisson (Spatial CV)	3.54	5.95	36.38	59.82	36.30	74.87
Bayesian (brms)						
Hurdle NegBinomial	5.00	3.67	43.41	47.99	—	—
NegBinomial	3.75	4.92	50.52	48.18	—	—

Table 2: Observed plant counts per 3×3 m quadrat on occupied plots (count > 0) in each area, giving the scale against which the mean absolute errors in the preceding table should be read.

Area	occ. n	25th pct	median	75th pct	max
High elevation (interpolation)	60	3	6	7	30
Low: Cochetopa (interpolation)	20	31	56	108	156
Low: Antero (extrapolation)	16	32	65	108	171